

Euseok Han

Bachelor of Science

Electrical Engineering (Major) & Computer Science (Double Major)

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RESEARCH INTERESTS

• Robot learning, 3D scene understanding, autonomous driving; broadly, AI that understands the physical world

EDUCATION

• **Korea Advanced Institute of Science and Technology, Daejeon**

Mar. 2020 – Aug. 2027 (Expected)

B.S. in Electrical Engineering and Computer Science

GPA: 4.02/4.3

• **University of Texas at Austin**

Aug. 2025 – Dec. 2025

Exchange Program

GPA: 4.0/4.0

PUBLICATIONS

[1] **SenseFuse: Label-Free Fusion of Image and Shape Encoders for Open-Vocabulary 3D Instance Segmentation** *Under review*

Euseok Han, Tri Ton, Hwanhee Kim, Seungyeon Ryu, Chang D. Yoo

ICRA 2027

[2] **Hamiltonian Invariant-Driven Mode-Transition IMM for Satellite Maneuver Detection**

Manuscript in preparation

Euseok Han, Sangheon Choi, Seung-Hyun Kong

RESEARCH EXPERIENCE

• **Artificial Intelligence Machine Learning Lab, KAIST**

Jan. 2025 – Feb. 2025, Jan. 2026 – Present

Undergraduate Researcher / Advisor: Prof. Chang Dong Yoo

Daejeon, KR

– URP: *From Point Cloud to Actions: 3D Structural Keypoint-Conditioned Hierarchical Robot Policy Learning*

– Fused frozen 2D image and 3D shape encoders at mask labeling, exploiting their largely disjoint failure patterns [1]

– Closed-form, label-free scene-level weight calibrated in 3.5 ms; median +7.1 pts across three benchmarks

• **NC Lab, KAIST**

Mar. 2025 – Aug. 2025

Undergraduate Researcher / Advisor: Prof. June-hwa Song

Daejeon, KR

– *Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing*

– Participated in experiment design and conducted a preliminary experiment using Zigbee communication modules

• **Control and Intelligent Systems Lab, KAIST**

Jul. 2024 – Aug. 2024

Undergraduate Researcher / Advisor: Prof. Heejin Ahn

Daejeon, KR

– Reproduced *Constrained Policy Optimization (ICML 2017)* baselines in MuJoCo from the official code (safe RL)

SELECTED PROJECTS

• **LLM-Guided Autonomous Robots**

Mar. 2026 – Jun. 2026

EE40078 Introduction to Multi-disciplinary Robotics, School of Electrical Engineering

– LLM-guided autonomous mobile delivery robot; RTAB-Map localization, path planning, and object pick-up

• **Hamiltonian Invariant-Driven Mode-Transition IMM for Satellite Maneuver Detection**

Mar. 2026 – Present

MO50125 The Principles and Applications of the Kalman Filter, Cho Chun Shik Graduate School of Mobility

– Satellite maneuver detection; IMM transition probabilities driven by a CUSUM on the J_2 -Hamiltonian Lie residual [2]

• **PintOS-KAIST**

Sep. 2024 – Dec. 2024

CS330 Operating Systems, School of Computing

– Implemented kernel modules for process scheduling, syscall, memory management of an educational x86-64 OS

OTHER EXPERIENCE

•**Republic of Korea Air Force (mandatory military service)**

Apr. 2022 – Jan. 2024

Information System Operator, Staff Sergeant, KAMD Operations Center

Osan Air Base, KR

– Operated real-time data processing servers for the Korea Air and Missile Defense (KAMD) center; 24/7 availability

TECHNICAL SKILLS AND INTERESTS

Programming Languages: C, C++, Python

Libraries: ROS, NumPy, PyTorch

Miscellaneous tools: MATLAB, L^AT_EX, Docker

Extracurricular: STROKE (Tennis Club)